PRANJAL KARAN

Computer Science Engineering Student

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Education

VIT Bhopal University – Bachelor of Technology, Computer Science & Engineering — Sehore, MP **Sept 2023 – May 2027**

CGPA: 8.70/10

Technical Skills

Languages: JavaScript (ES6+), TypeScript, C++, Java

Backend & APIs: Node.js, Express.js, RESTful API Design, JWT Authentication, Clerk Auth

Frontend: React, HTML5, CSS3, Tailwind CSS v4, Vite, Recharts

Databases: MongoDB, MySQL, PostgreSQL, SQL

Web & Networking: HTTP/HTTPS, DNS, TLS, CORS, REST APIs, Network Debugging

Engineering & DevOps: Git, GitHub, Automated Testing, Vitest, Supertest, Pino Logging, Rate Limiting

Integrations & AI: Google Gemini API, Ollama, Razorpay APIs, LLM API Integration, API Data Synchronization

Projects

ReconAI – Verification-First AI Finance Controller Platform — GitHub **2026** React, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, Zod, Vitest, Gemini, Razorpay

- **Built** a full-stack financial operations platform that reconciles merchant orders, gateway payments, and settlement records using an **integer-paise deterministic engine**, processing a **120-scenario benchmark** with **100% classification accuracy**.
- **Engineered** an **autonomous 9-state AI Finance Controller Agent** that orchestrates multi-pass candidate matching, confidence safety gates, exception creation, automated AI investigation dispatch, and batch report synthesis.
- **Integrated Razorpay Test Mode** as a read-only API source with PII stripping, idempotent bulk upserts, and live-key guards; added **Google Gemini** for advisory root-cause analysis with a deterministic fallback engine.
- **Verified** system reliability with **181 automated backend tests across 42 test files** and a passing Vite production build, enforcing append-only audit trails, credential redaction, and strict financial safety boundaries.

HealthPulse – Full-Stack Healthcare Appointment & Clinical Platform — Live — GitHub **2026** React 18, TypeScript, Node.js, Express.js, MongoDB, JWT, Google Gemini, Ollama

- **Architected** a **40,000+ line** full-stack platform with **61 REST API endpoints** across 9 modules and **13 MongoDB schemas**, following a strict controller-service-model SDLC architecture.
- **Eliminated** double-booking race conditions under concurrent load by combining a **database-level compound partial unique index** with **TTL-based 5-minute slot holds**, guaranteeing atomic data integrity at the database engine tier.
- **Engineered** stateless JWT authentication with a `tokenVersion` mechanism enabling instant session revocation, role-based access control across 3 personas, and rate-limited endpoints to prevent brute-force attacks.
- **Orchestrated** a dual-engine AI clinical-summary pipeline (Google Gemini + Ollama via `Promise.allSettled`) with a zero-hallucination medication-verification guard, backed by **154 automated tests (471 assertions)** and a **98/100** Lighthouse performance score.

Achievements & Certifications

Achievements:

- **National Qualifier** – Cleared Round 1 of **TCS CodeVita**, a national-level competitive coding challenge
- **Co-Lead (Social Media)**, Linpack Club, VIT Bhopal – Increased event engagement by **30%** through targeted digital campaigns

Certifications:

- **Microsoft Certified: Azure AI Cloud Developer Associate** (Score: 922/1000 — ID: DDFC0151CEFD415C)
- **Microsoft Certified: Azure Data Fundamentals** (Score: 955/1000)
- **MERN Full Stack Certification** (Ethnus & Codemithra)